

Game Theory Ten

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Contents

1	BNN Error Orders	2
1.1	Results with Fixed Code	2
2	From Navier–Stokes to Mean Field Games	2
2.1	Derivation from Newton’s Second Law	2
2.2	Conservation of Mass	3
2.3	Compressible Navier–Stokes System	3
2.4	Relation to Incompressible Navier–Stokes	3
3	Luo and Song’s Variation of Compressible Navier–Stokes	4
3.1	Luo and Song’s Main Result	4
4	Extension to HJB Equations	5
5	Hamilton–Jacobi–Bellman Equations	5
5.1	Deterministic HJB Equation	5
5.2	Stochastic HJB Equation	6
6	How Luo and Song’s PDE Becomes an HJB Equation	6
7	How the HJB Equation Relates to Physics	7
7.1	Least Action Principle	7
7.2	Compressible Fluid Lagrangian	7
8	From Luo and Song to Mean Field Games	8
9	Interpretation of the Mean Field Games PDE	8
10	Fokker–Planck Equation and the MFG Density PDE	9
11	Coupled Mean Field Games Picture	9

1. BNN Error Orders

Last week's numerical experiments modeled two player, two strategy games with the following initial conditions:

$$x^1(0) = \frac{1}{2}, \quad y^1(0) = \frac{1}{2},$$

running from

$$T = 0 \quad \text{to} \quad T = 10.$$

After looking back through my code, I identified two issue in the original implementation:

- a missing factor of 2 in the ODE,
- the actual step size varying from the expected step size because of `np.linspace()`.

1.1. Results with Fixed Code

After fixing by BNN ODE code, I found fourth-order convergence for both the one and two strictly dominant strategy case. In both of these cases, the Jacobian of the ODE never changed, due to the strict dominance and initial starting condition.

In the one strictly dominant strategy case, with the provided initial conditions, I found that the exact solution was:

$$\begin{aligned} x_1(t) &= \frac{1}{2+2t}, & y_1(t) &= \frac{1+2t}{2+2t} \\ x_2(t) &= \frac{1}{2-2\ln(1+t)+2t}, & y_2(t) &= \frac{1-2\ln(1+t)+2t}{2-2\ln(1+t)+2t} \end{aligned}$$

However, in the case where neither player had a strictly dominant strategy, the convergence appeared to be roughly second order and noisy. The likely case is the domain dependent Jacobian, with non-differentiable transitions at $x^1 = \frac{1}{2}$ and $x^2 = \frac{1}{2}$. I was unable to find an exact solution so the "actual solution" was computed by an adaptive Runge-Kutta method.

2. From Navier–Stokes to Mean Field Games

2.1. Derivation from Newton's Second Law

Newton's second law says:

$$\text{Force} = \text{Mass} \times \text{Acceleration}.$$

For a fluid of density ρ and velocity v , this is written as

$$\rho \partial_t v = \text{sum of forces}.$$

The forces identified in the slides are:

$$\begin{aligned}\text{Inertial force} &: -\rho v \cdot \nabla v, \\ \text{Net pressure} &: -\nabla p, \\ \text{Viscosity} &: \mu \Delta v, \\ \text{External forces} &: f.\end{aligned}$$

Combining these gives

$$(\partial_t + v \cdot \nabla)v = \frac{-\nabla p + \mu \Delta v + f}{\rho}.$$

2.2. Conservation of Mass

The change in density is $\partial_t \rho$, and the mass flux term is

$$-\nabla \cdot (\rho v).$$

Thus conservation of mass becomes

$$\partial_t \rho + \nabla \cdot (\rho v) = 0.$$

2.3. Compressible Navier–Stokes System

Putting the equations together gives

$$\begin{aligned}(\partial_t + v \cdot \nabla)v &= \frac{-\nabla p + \mu \Delta v + f}{\rho}, \\ \partial_t \rho + \nabla \cdot (\rho v) &= 0,\end{aligned}$$

with initial data

$$v(0, x) = v_0(x), \quad \rho(0, x) = \rho_0(x).$$

The variables are:

$$v = \text{velocity}, \quad \rho = \text{density}, \quad p = \text{pressure}, \quad \mu = \text{viscosity}, \quad f = \text{forcing function}.$$

2.4. Relation to Incompressible Navier–Stokes

If ρ is constant, then the compressible equations reduce to the incompressible equations:

$$(\partial_t + v \cdot \nabla)v = \nu \Delta v - \nabla p + f,$$

$$\nabla \cdot v = 0.$$

In this case, the density scaling is absorbed into the pressure, viscosity ($\nu = \frac{\mu}{\rho}$), and forcing function.

3. Luo and Song's Variation of Compressible Navier–Stokes

Luo and Song introduced three changes to the compressible Navier–Stokes equation:

1. The ratio

$$\frac{-\nabla p + \mu \Delta v + f}{\rho}$$

was generalized to the functional

$$-\nabla p[\rho] + \mu[\rho] \Delta v + f[\rho].$$

2. The continuity equation was replaced by a Fokker–Planck-type equation:

$$\partial_t \rho + \nabla \cdot (\rho v) + \mu[\rho] \Delta \rho = 0.$$

3. A terminal condition was imposed on the density instead of an initial condition.

This yields the generalized system

$$(\partial_t + v \cdot \nabla) v = -\nabla p[\rho] + \mu[\rho] \Delta v + f[\rho],$$

$$\partial_t \rho + \nabla \cdot (\rho v) + \mu[\rho] \Delta \rho = 0,$$

$$v(0, x) = v_0(x), \quad \rho(T, x) = \rho_T(x).$$

3.1. Luo and Song's Main Result

Luo and Song then made the following assumptions:

$$\mu[\rho] = \frac{1}{2}, \quad f[\rho] = 0, \quad v_0(x) = \nabla h(x),$$

where h is a smooth function.

Under these assumptions, the system becomes

$$(\partial_t + v \cdot \nabla) v = -\nabla p[\rho] + \frac{1}{2} \Delta v,$$

$$\partial_t \rho + \nabla \cdot (\rho v) + \frac{1}{2} \Delta \rho = 0,$$

$$v(0, x) = \nabla h[\rho](x), \quad \rho(T, x) = \rho_T(x).$$

Under additional hypotheses on the domain (3-torus) and suitable norms on p and h , Luo and Song prove existence of a classical solution to this set of PDEs.

4. Extension to HJB Equations

The velocity and density equations can be rewritten as:

$$\left(\partial_t + v \cdot \nabla - \frac{1}{2}\Delta\right)v = -\nabla p[\rho], \quad v(0, x) = \nabla h[\rho](x),$$

and

$$\left(\partial_t - \frac{1}{2}\Delta\right)\rho - \nabla \cdot (\rho v) = 0, \quad \rho(T, x) = \rho_T(x).$$

Now define

$$v = \nabla u.$$

Then

$$\left(\partial_t + v \cdot \nabla - \frac{1}{2}\Delta\right)v = \partial_t \nabla u + (\nabla u \cdot \nabla) \nabla u - \frac{1}{2}\Delta \nabla u.$$

Using

$$(\nabla u \cdot \nabla) \nabla u = \nabla \left(\frac{1}{2} |\nabla u|^2 \right),$$

we get

$$\left(\partial_t + v \cdot \nabla - \frac{1}{2}\Delta\right)v = \nabla \left(\partial_t u + \frac{1}{2} |\nabla u|^2 - \frac{1}{2} \Delta u \right).$$

Thus

$$\nabla \left(\partial_t u + \frac{1}{2} |\nabla u|^2 - \frac{1}{2} \Delta u \right) = -\nabla p[\rho].$$

Ignoring additive constants, this yields the HJB-type equation

$$\partial_t u + \frac{1}{2} |\nabla u|^2 - \frac{1}{2} \Delta u = -p[\rho],$$

with

$$u(0, x) = h[\rho](x).$$

5. Hamilton–Jacobi–Bellman Equations

5.1. Deterministic HJB Equation

Consider the objective functional

$$J(x_0, t_0, a) = \int_{t_0}^T I(\tau, x(\tau), a(\tau)) d\tau + C(x(T)),$$

where

$$x(t_0) = x_0, \quad \dot{x}(t) = g(t, x(t), a(t)).$$

The value function is defined as

$$V(x, t) = \sup_{a(\cdot)} J(x, t, a).$$

The deterministic HJB equation for this system is

$$-\frac{\partial V}{\partial t} = \sup_a \left[I(t, x, a) + \nabla V \cdot g(t, x, a) \right],$$

with terminal condition

$$V(x, T) = C(x).$$

5.2. Stochastic HJB Equation

For the stochastic system

$$dX(t) = g(t, X(t), a(t)) dt + \sigma(X(t)) dW(t),$$

the value function becomes

$$V(x, t) = \sup_{a(\cdot)} \mathbb{E}[J(x, t, a)].$$

Assuming isotropic diffusion, the stochastic HJB equation is

$$-\frac{\partial V}{\partial t} = \sup_a \left[I(t, x, a) + \nabla V \cdot g(t, x, a) \right] + \frac{1}{2} \sigma(x)^2 \Delta V,$$

with

$$V(x, T) = C(x).$$

6. How Luo and Song's PDE Becomes an HJB Equation

To match Luo and Song's equation with stochastic HJB, we assume

$$g(t, x, a) = a, \quad \sigma(x) = 1, \quad I(t, x, a) = -\frac{1}{2}|a|^2 + p[\rho].$$

Then

$$\sup_a \left[I(t, x, a) + \nabla V \cdot g(t, x, a) \right] = \sup_a \left[\nabla V \cdot a - \frac{1}{2}|a|^2 \right] + p[\rho].$$

The supremum is attained at

$$a = \nabla V,$$

and its value is

$$\sup_a \left[\nabla V \cdot a - \frac{1}{2}|a|^2 \right] = \frac{1}{2} |\nabla V|^2.$$

Hence

$$-\frac{\partial V}{\partial t} = \frac{1}{2} |\nabla V|^2 + p[\rho] + \frac{1}{2} \Delta V,$$

or equivalently

$$\partial_t V + \frac{1}{2} |\nabla V|^2 + \frac{1}{2} \Delta V = -p[\rho].$$

Now set

$$V = -u, \quad t \mapsto T - t, \quad C(x) = -h[\rho](x).$$

This gives Luo and Song's equation in the form

$$\partial_t u + \frac{1}{2}|\nabla u|^2 - \frac{1}{2}\Delta u = -p[\rho], \quad u(0, x) = h[\rho](x).$$

7. How the HJB Equation Relates to Physics

7.1. Least Action Principle

The action functional is

$$S[x(\cdot)] = \int_0^T \mathcal{L}(x, \dot{x}, t) dt.$$

The least action principle says that nature selects x to minimize the action.

This leads to the Euler–Lagrange equation:

$$\frac{d}{dt} \left[\frac{\partial \mathcal{L}}{\partial \dot{x}} \right] = \frac{\partial \mathcal{L}}{\partial x}.$$

The Hamiltonian is constructed by

$$H = \sup_{\dot{x}} [p \cdot \dot{x} - \mathcal{L}],$$

where the conjugate momentum is

$$p = \frac{\partial \mathcal{L}}{\partial \dot{x}}.$$

The HJB equation may then be viewed locally as

$$\frac{\partial V}{\partial t} = -H(x, p, t),$$

so HJB may be interpreted as a local version of the least action principle.

7.2. Compressible Fluid Lagrangian

The Lagrangian corresponding to a compressible fluid is

$$\mathcal{L}(\rho, u) = \frac{1}{2}\rho|\nabla u|^2 - p[\rho].$$

The conjugate momentum is

$$p = \frac{\partial \mathcal{L}}{\partial(\nabla u)} = \rho \nabla u.$$

The associated Hamiltonian is written as

$$H = \sup_u [p \cdot u - \mathcal{L}] = \frac{1}{2}\rho|\nabla u|^2 + p[\rho].$$

This leads formally to

$$\frac{\partial u}{\partial t} = -H,$$

that is,

$$\frac{\partial u}{\partial t} + \frac{1}{2}\rho|\nabla u|^2 = -p[\rho].$$

This is the deterministic form of Luo and Song's equation

8. From Luo and Song to Mean Field Games

A representative mean field games system in the slides is

$$\partial_t u - \nu \Delta u + H(x, \nabla u) = W(m), \quad u(x, 0) = W_0(m(0)),$$

$$\partial_t m + \nu \Delta m + \nabla \cdot \left(\frac{\partial H}{\partial p} m \right) = 0, \quad m(x, T) = m_0(x).$$

We can compare this with the Luo–Song system

$$\partial_t u - \frac{1}{2}\Delta u + \frac{1}{2}|\nabla u|^2 = -p[\rho], \quad u(0, x) = h[\rho](x),$$

$$\partial_t \rho + \frac{1}{2}\Delta \rho + \nabla \cdot (\rho \nabla u) = 0, \quad \rho(T, x) = \rho_T(x).$$

The Luo–Song PDE can be viewed as a mean field game with

$$\nu = \frac{1}{2}, \quad H(x, \nabla u) = \frac{1}{2}|\nabla u|^2.$$

9. Interpretation of the Mean Field Games PDE

In the mean field games limit, players are assumed to be indistinguishable. Instead of each player having a separate value function, there exists a common value function

$$u(x, t),$$

depending only on the state x and time t . This function tells you the value of playing strategy x at time t .

The stochastic HJB equation associated with Mean Field Games is

$$\partial_t u - \nu \Delta u + H(x, \nabla u) = W(m), \quad u(x, 0) = W_0(m(0)).$$

We can interpret the terms as follows:

- $\nu \Delta u$ represents diffusion of player strategies due to randomness;
- $H(x, \nabla u)$ encodes best response of each player;
- $W(m)$ is the interaction cost coming from the other agents.

A popular example is

$$I(t, x, a) = -\frac{1}{2}|a|^2, \quad g(t, x, a) = a,$$

for which

$$H(x, \nabla u) = \sup_a \left[\nabla u \cdot a - \frac{1}{2} |a|^2 \right] = \frac{1}{2} |\nabla u|^2.$$

10. Fokker–Planck Equation and the MFG Density PDE

If a stochastic process satisfies

$$dX_t = \mu(X_t, t) dt + \sigma dW_t,$$

then its density $\rho(x, t)$ satisfies the Fokker–Planck equation

$$\partial_t \rho = -\nabla \cdot (\mu \rho) + \frac{\sigma^2}{2} \Delta \rho.$$

The mean field games density equation is

$$\partial_t m + \nu \Delta m + \nabla \cdot \left(\frac{\partial H}{\partial p} m \right) = 0, \quad m(x, T) = m_0(x).$$

This is a Fokker–Planck equation with reverse time evolution, with

$$\sigma = 1, \quad \mu(X_t, t) = -\frac{\partial H}{\partial p}.$$

The terms are interpreted as:

- $\nu \Delta m$: diffusion of player strategies due to randomness,
- $\frac{\partial H}{\partial p} m$: drift of player strategies as they move toward optimal strategies.

11. Coupled Mean Field Games Picture

The final interpretation is that:

- the HJB equation describes how an individual optimally reacts to the population,
- the Fokker–Planck equation describes how the population moves in response to those optimal individual reactions.

Thus the system

$$\begin{aligned} \partial_t u - \nu \Delta u + H(x, \nabla u) &= W(m), \\ \partial_t m + \nu \Delta m + \nabla \cdot \left(\frac{\partial H}{\partial p} m \right) &= 0 \end{aligned}$$

captures the two-way interaction between optimal decision-making and aggregate evolution.